



Field Station Fundamentals

Session 2 — Radio, Antennas & Power

TavaOne Education · W4GGJ



Today's Session

- Radio selection — which rig for portable HF
- Antennas — EFHW, dipoles, verticals, deployment tools
- Feedline and connectors — what to carry
- Power — LiFePO4 batteries, capacity math, solar
- Building your go-kit — checklist and weight targets
- **Hands-on: set up the club IC-705 + EFHW on the patio**



Radio Selection

What to bring — and why it matters less than you think



POTA Radio — What You Need

- All-band HF transceiver (at minimum 40m + 20m)
- Portable enough to carry to an operating position
- Built-in or external antenna tuner
- Reasonable battery draw for your expected activation length

- Popular choices (ascending price):
 - Xiegu G90 — 20W, 1.8 lb, built-in tuner and SDR, ~\$450
 - Xiegu G106 — 10W, 0.9 lb, touchscreen, ~\$350
 - Yaesu FT-891 — 100W, 2.2 lb, full power, needs external battery, ~\$600
 - Icom IC-705 — 10W, HF+VHF+UHF, built-in battery, GPS, ~\$1,350
 - Elecraft KX2 — 10W, 0.7 lb, AA batteries, keyer, ~\$950



Power Considerations — TX Current

- Higher power = easier contacts, shorter battery life
- Typical TX current draw (at 12V):
 - 10W radio: 2–4A TX
 - 50W radio: ~10–12A TX
 - 100W radio: 15–22A TX
- RX draw is usually 1–2A regardless of power level
- SSB duty cycle: ~30% transmit / 70% receive
- FT8 duty cycle: ~50% transmit
- CW duty cycle: ~40% (depends on speed and spacing)



Antennas for Portable Ops

The wire that does the work



End-Fed Half-Wave (EFHW) — Why It Won

- The most popular POTA antenna — and for good reason
- How it works:
 - A half-wave wire fed at the high-impedance end
 - 49:1 transformer drops impedance to match 50-ohm coax
 - 40m wire: 66 feet — resonant on 40m, 20m, 15m, 10m (harmonics)
- Why activators love it:
 - Only needs support at ONE end (tree, squid pole)
 - No ground plane required
 - 4 bands, no tuner adjustment between them
 - Entire kit weighs under 1 lb and fits in a sandwich bag
- Build cost: \$15–20 in parts (FT140-43 toroid + wire)



Wire Dipole vs. Linked Dipole



Center-Fed Wire Dipole

- Half-wave per side (33 ft per leg on 40m)
- 50-ohm match — excellent SWR without a tuner
- Needs 3 support points (center + 2 ends)
- One band per build (need separate dipoles per band)
- Works well as inverted-V with center up
- Easy to build: center insulator + 2 wires + feedline



Linked Dipole

- Quick-disconnect links (bullet connectors) in each leg
- Remove links to shorten = change bands
- No tuner needed for each resonant band
- Common builds: 40/20/15/10m (3 links per side)
- More deployment steps — clip links before TX
- Popular with CW operators who hate tuner losses



Verticals and Deployment Tools

- Verticals — when trees aren't available:
 - Chameleon MPAS Lite: versatile, needs radials, ~\$200
 - Wolf River Coil: loaded vertical, compact, ~\$80
 - Commercial hamsticks on a mag mount: vehicle-portable
 - Key: always include a counterpoise or radials — skip it and efficiency tanks
- Deployment tools:
 - Throw bag + paracord: 3 oz bag + 100 ft line; toss over a branch
 - Squid pole (fiberglass telescoping): 7–10m up to 60cm collapsed; best all-around
 - Collapsed fishing rod: cheap substitute, works fine
 - Painter pole: heavy but inexpensive and common
- Practice throw-bag technique at home before your first activation



Power Sources

LiFePO₄, solar, and the capacity math



LiFePO₄ — The Standard

- Lithium Iron Phosphate: the POTA battery of choice
- Why LiFePO₄ beats lead-acid and standard lithium:
 - Safe chemistry — won't catch fire if punctured or overcharged
 - Flat discharge curve — stays at 13.2V until nearly empty
 - 1C discharge rate — 20Ah battery delivers 20A continuously
 - 2000+ charge cycles (lead-acid: ~500)
 - Lighter: ~60% the weight of equivalent AGM
- Popular brands: Bioenno, Dakota Lithium, Battleborn, Renogy
- Do NOT discharge below 20% — protect cycle life
- Charge with a LiFePO₄-compatible charger — not a car charger



Capacity Math — Size Your Battery

- Formula: Average draw = (TX amps × TX duty%) + (RX amps × RX duty%)
- Example — FT-891 at 50W, SSB:
 - TX draw at 50W: 12A
 - RX draw: 1.5A
 - TX duty: 30%
 - Average = $(12 \times 0.3) + (1.5 \times 0.7) = 3.6 + 1.05 = 4.65\text{A}$
 - 20Ah battery usable (80%) = 16Ah
 - Runtime: $16 \div 4.65 = 3.4$ hours
- Example — IC-705 at 10W, FT8:
 - TX draw: 3A | RX: 1A | TX duty: 50%
 - Average = $(3 \times 0.5) + (1 \times 0.5) = 2\text{A}$
 - 10Ah battery (8Ah usable) $\div 2\text{A} = 4$ hours



Solar and USB Power

- Solar — supplemental, not primary for most activations
 - 40W panel in full Florida sun: ~3A charge rate at 12V
 - Pair with MPPT charge controller (Renogy 10A, ~\$25)
 - Weight penalty: ~2.5 lbs for 40W panel
 - Best use: vehicle-portable multi-park day, long activations
 - Not worth it for a single 90-minute activation — just bring a bigger battery
- Generators — last resort
 - Heavy, noisy, need fuel, many parks prohibit them
 - If you think you need a generator, you've over-estimated power needs
 - Solution: run lower power instead
- USB power bank for logging devices
 - Keep phone/tablet charged separately from the radio battery
 - 10,000–20,000mAh bank handles a full activation day



Building Your Go-Kit

Ready to grab, ready to go



The Go-Kit Philosophy

- Should be ready to grab at any moment
- Activation opportunity appears → you're on the road in 10 minutes
- Everything lives in one bag, always stocked
- Essentials:
 - Radio + power cable + mic/headphones
 - Antenna (EFHW or dipole) + feedline + adapters
 - Battery (charged)
 - Throw bag + line, or squid pole
 - Logging device + paper backup + pens
 - USB power bank
 - FCC license (printed or phone)
- Comfort:



Weight Targets by Setup Type



Vehicle-Portable

- Park within walking distance of car
- Under 40 lbs total — anything goes
- Full station: 100W radio, big battery, real antenna
- Comfort items: table, chair, umbrella
- Best for first activations — minimize variables



Hike-Portable

- Short hike (<1 mile): under 25 lbs
- Long hike / backpack: under 10 lbs
- QRP radio (KX2, IC-705, G106)
- 66 ft EFHW on a squid pole
- 10Ah LiFePO4, phone logging
- Everything fits in a daypack

Station Setup — Bag to RF-Ready

Team roles (everyone rotates):

- Radio operator — unpack, connect power + feedline
- Antenna crew — unwind EFHW, connect to feedline, deploy
- Power crew — check battery voltage, connect to radio
- Logger — open HAMRS, start activation, enter park reference **Done when: radio powered, antenna up, SWR checked, HAMRS loaded** Target: RF-ready in under 15 minutes



KNOWLEDGE CHECK

A 40m EFHW is built with 66 feet of wire. Without any changes, it also resonates on which bands?

A. 80m and 160m (longer harmonics)

B. 20m, 15m, and 10m (harmonic multiples)

C. Only 40m — EFHW is single-band



KNOWLEDGE CHECK

Why is LiFePO₄ preferred over lead-acid (AGM/SLA) for portable POTA operations?

A. LiFePO₄ is cheaper per Ah

B. LiFePO₄ is lighter, has more cycles, and holds voltage better through discharge

C. LiFePO₄ doesn't need a BMS or charge controller



KNOWLEDGE CHECK

You arrive at an open field with no trees. Your antenna kit contains an EFHW, a throw bag with paracord, and an 8m squid pole. What do you do?

A. Give up and drive to a park with trees

B. Deploy the EFHW vertically using the squid pole as a mast

C. The EFHW won't work without a tree — need a different antenna



Session 3 Preview

Operating, Logging & Uploading

- CQ POTA calling procedure and exchange
 - Self-spotting on pota.app — getting hunters on your frequency
 - FT8 from the field — WSJT-X setup and the digital workflow
 - HAMRS logging — walkthrough from activation create to ADIF export
 - Log upload to pota.app — full workflow, common errors
 - **Simulated activation: log 10 contacts start-to-finish**
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